

Lily Koffman

BIostatISTICS PHD CANDIDATE · JOHNS HOPKINS BLOOMBERG SCHOOL OF PUBLIC HEALTH

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Education

Johns Hopkins Bloomberg School of Public Health

Baltimore, MD

PHD CANDIDATE IN BIostatISTICS

2021 - present

- Advisors: Dr. Ciprian Crainiceanu and Dr. John Muschelli
- Dissertation defense: March 24, 2026
- Dissertation: Statistical Methods for Digital Fingerprinting and Health Prediction Using High-Resolution Accelerometry Data

Harvard TH Chan School of Public Health

Boston, MA

MS BIostatISTICS

2019-2021

- Research advisor: Dr. Sebastien Haneuse

Harvard University

Cambridge, MA

AB STATISTICS

2015-2019

- Thesis research advisor: Kevin Rader

Publications

PUBLISHED & PEER REVIEWED

- Goeddel, L., **Koffman, L.***, Zhou, X. et al. 2026. Low Cardiac Output During Periods of Hypotension and Risk of Acute Kidney Injury in Cardiac Surgery. *Accepted at British Journal of Anesthesia*.
- Crainiceanu, C., and **Koffman, L.** 2026. Discussion on “INTACT: A method for integration of longitudinal physical activity data from multiple sources” by Jingru Zhang, Erjia Cui, Hongzhe Li, and Haochang Shou. *Accepted at Biometrics*.
- Goeddel, L., Hernandez, M., **Koffman, L.**, et al. 2025. Assessment of Renal Vein Flow Index by Transesophageal Echocardiography: Precision, Variability, and Association with Cardiac Index During Cardiac Surgery. *Journal of Cardiothoracic and Vascular Anesthesia*
- Goeddel, L., Hernandez, M., **Koffman, L.**, et al. 2025. Fine-Mapping the Association of Acute Kidney Injury with Mean Arterial and Central Venous Pressures during Coronary Artery Bypass Surgery. *Anesthesia & Analgesia*
- Gao, S., Zhou, X., **Koffman, L.** et al. 2025. Comparing step counting algorithms for high-resolution wrist accelerometry data in older adults in the ARIC study. *The Journals of Gerontology: Series A*.
- Koffman, L.**, Crainiceanu, C.M., Muschelli, J. 2024. Comparing Step Counting Algorithms for High-Resolution Wrist Accelerometry Data in NHANES 2011-2014. *Medicine & Science in Sport & Exercise*.
- Goeddel, L., Hernandez, M., **Koffman, L.** et al. 2024. Assessment of Renal Vein Stasis Index by Transesophageal Echocardiography During Cardiac Surgery: A Feasibility Study. *Anesthesia & Analgesia*.
- Goeddel, L., **Koffman, L.**, Hernandez, M. et al. 2024. Occurrence of Low Cardiac Index During Normotensive Periods in Cardiac Surgery: A Prospective Cohort Study Using Continuous Noninvasive Cardiac Output Monitoring. *Anesthesia & Analgesia*.
- Koffman, L.**, Muschelli, J. 2024. Evaluating Step Counting Algorithms on Subsecond Wrist-Worn Accelerometry: A Comparison Using Publicly Available Data Sets. *Journal for the Measurement of Physical Behaviour*.
- Koffman, L.**, Crainiceanu, C.M., Leroux, A. 2024. Walking fingerprinting. *Journal of the Royal Statistical Society Series C: Applied Statistics*.
- Koffman, L.**, Crainiceanu, C.M., Roemmich, R.T., French, M.A. 2023. Identifying Unique Subgroups of Individuals With Stroke Using Heart Rate and Steps to Characterize Physical Activity. *Journal of the American Heart Association*.
- Lin J. J. Y., **Koffman, L.**, Tehrani, M. W., et al. 2023. Reliability of low mass toenail samples as biomarkers of chronic metal exposure. *Journal of Exposure Science and Environmental Epidemiology*.

- Koffman, L.**, Zhang, Y., Harezlak, J., Crainiceanu, C.M., Leroux, A. 2023. Fingerprinting walking using wrist-worn accelerometers. *Gait & Posture*.
- Koffman, L.**, Levis, A.W., Haneuse, S. et al. 2021. Evaluation of Intensive Telephonic Nutritional and Lifestyle Counseling to Enhance Outcomes of Bariatric Surgery. *Obesity Surgery*.
- Koffman, L.**, Levis, A.W., Arterburn, D. et al 2021. Investigating Bias from Missing Data in an Electronic Health Records-Based Study of Weight Loss After Bariatric Surgery. *Obesity Surgery*.

UNDER REVIEW

- Koffman, L.**, Muschelli, J., Crainiceanu, C.M. 2025. Walking Fingerprinting Using Wrist Accelerometry During Activities of Daily Living in NHANES. *Invited to resubmit at Annals of Applied Statistics* [arXiv:2506.17160](https://arxiv.org/abs/2506.17160)
- Koffman, L.**, Gao, S., Zhou, X., Leroux, A., Crainiceanu, C.M, Muschelli, J. 2025. Function on Scalar Regression with Complex Survey Designs. *Submitted to Biometrics* <https://arxiv.org/abs/2511.05487>
- Brantner, C., **Koffman, L.**, Freemas, J., Stoms, M., Goldsmith, J. 2025. Making Sense of the Cycle: Statistical Modeling Techniques for Menstrual Health Data. *Submitted to Women's Reproductive Health*

* denotes co-first author

Software and Datasets

- `svyfosr`: Survey-Weighted Function on Scalar Regression. **Koffman, L.** and Muschelli, J. *Submitted to CRAN*.
- `accelPrint`: Fingerprinting for acceleration and walking. **Koffman, L.** and Muschelli, J. *GitHub*.
- `bloodlines`: Generate nice plots for hemodynamics data. **Koffman, L.** *GitHub*.
- Minute level step counts and physical activity data from the National Health and Nutrition Examination Survey (NHANES) 2011-2014. **Koffman, L.** and Muschelli, J. *PhysioNet*.

Talks and Posters

CONTRIBUTED POSTERS

- Koffman, L.**, Adamowicz, L., Moustridi, E. et al. 2025. Comparing Literature Algorithms to Derive Physical Activity Endpoints from a Wrist Accelerometer. Poster: IEEE BSN, Los Angeles, CA.
- Koffman, L.**, Muschelli, J., Crainiceanu, C. 2025. Fingerprinting walking in a large epidemiological study. Poster: ENAR, New Orleans, LA.
- Koffman, L.**, Crainiceanu, C., Roemmich, R., French, M.A. 2023. Identifying Unique Subgroups of Individuals Post-Stroke Using Heart Rate and Steps to Characterize Physical Activity. Poster: APTA-CSM, San Diego, CA.
- Balasubramanian, A., French, M., **Koffman, L.**, et al. 2023. Classifying Physical Conditioning Using Remote Monitoring in Chronic Obstructive Pulmonary Disease. Poster. American Thoracic Society International Conference, Washington D.C.

TALKS

- Walking Fingerprinting Using Wrist Accelerometry during Activities of Daily Living in NHANES*. ENAR Spring Meeting, March 2026
- Walking Fingerprinting in a large epidemiological study*. Building Future Faculty Workshop, NC State University, April 10, 2025* and Functional Data Working Group at Emory (FUDGE), April 2025*
- NHANES Accelerometry and Derivatives*. BIRS: Emerging Statistical Methods for Digital Health Data, February 2025
- Walking Fingerprinting*. ENAR Spring Meeting, March 2024
- Application of Open-Source Step Counting Algorithms on Publicly-Available Data*. Pre-ENAR Workshop: Statistical Methods for Digital Health Technologies Data, March 2024
- A nonrandom sample of medical statistics*. MedStar Health General Surgery Baltimore Grand Rounds, March 2024*
- Living On the Edge: U.S. Ski & Snowboard's Competitive Edge via Analytics*. Panelist: Comcast Sports Tech Webinar, Dec 2020*

*denotes invited talk

Professional Experience

Pfizer

Cambridge, MA

STATISTICS INTERN

Summer 2024

- AI/ML Quantitative and Digital Sciences Team
- Mentor: Nunzio Camerlingo
- Analysis of treatment effects using data collected from wearable digital health technologies

US Ski and Snowboard

Park City, UT

DATA & ANALYTICS INTERN WITH GUS KAEDING

2019-2020

- Prediction of alpine skiing performance from training data (article)
- Modeling of medal probabilities to inform funding decisions and team naming criteria
- Analysis of female participation in the American Birkebeiner
- Creation of app to visualize force plate output

Awards and Fellowships

- 2025-26 **Joseph Zeger Travel Award**, Johns Hopkins Bloomberg Department of Biostatistics
- 2019-2021 **Presidential Scholar**, Harvard TH Chan School of Public Health
- 2020 **WiST Fellowship**, Women in Sports Technology

Teaching

- 2023-26 **Statistical Methods in Public Health**, Lead Teaching Assistant
- Fall 2022 **Statistical Methods in Public Health**, Teaching Assistant

Lead Teaching Assistant • Competitive, appointed role involving leading multiple weekly lab sessions and overseeing course TA team.

Leadership

- 2023-2026 **Wearable and Implantable Technology (WIT) Research Group**, Organizer
- 2025 **JHU Biostatistics Writing Accountability Group (WAG)**, Founder & Organizer
- 2025 **Mobile and Wearable Data Science ASA Interest Group**, Communications Officer
- 2019 **Harvard University Division I Nordic Ski Team**, Captain